Sec: SC-60 **Qualitative Analysis** **Date: 08–07-2020**

SINGLE ANSWER TYPE

1. $MCl_3 \xrightarrow[\text{of water}]{\text{huge amount}} \text{white turbidity} \xrightarrow{AcOH}$ clear solution Then M^{3+} is

a) Fe^{3+} b) Sb^{3+} c) Bi^{3+} d) Au^{3+}

2. Choose the correct solubility order

a) $BaS < MnS < Sb_2S_3 < K_2S$ b) $K_2S < MnS < Sb_2S_3 < BaS$

c) $Sb_2S_3 < MnS < BaS < K_2S$ d) $Sb_2S_3 < BaS < MnS < K_2S$

3. When Ca^{2+}, Sr^{2+} and Ba^{2+} are taken together in a test tube and $Na_2C_2O_4$ solution is added, which of the following cations will forms the precipitated first?

a) Ca^{2+} b) Sr^{2+} c) Ba^{2+} d) All will form precipitated together

4. Na_2SO_4 solution $+M^{2+} \rightarrow$ yellow ppt. (A). The cation M^{2+} and yellow ppt. A are respectively

a) Sr^{2+} and $SrSO_4$ b) Hg^{2+} and $HgSO_4$

c) Hg^{2+} and $2HgO.HgSO_4$ d) Hg^{2+} and $HgO.2HgSO_4$

5. The precipitate of Ag_2S is not obtained from which of the following solutions awhen H_2S is passed through them?

a) Aqueous suspension of $AgCl$ b) $\left[Ag\left(S_2O_3\right)_2\right]^-$ solution.

c) $\left[Ag\left(CN\right)_2\right]^-$ solution d) $AgNO_3$ solution.

6. Aqueous solution of metal ion $M^{n+1} + KI \rightarrow$ Black ppt. $(N) \xrightarrow[\text{suspension is heated}]{\text{aqueous}}$ Orange ppt. (P) What is P?

a) $Bi(OH)_3$ b) Hg_2I_2 c) BiI_3 d) $BiO.I$

7. The aqueous solution of a salt having cation

$M^{2+} \xrightarrow{(NH_4)_2S}$ ppt. Which is soluble in acetic acid Then cation M^{2+} is

a) Mn^{2+} b) Zn^{2+} c) Ni^{2+} d) Hg^{2+}

MULTIPLE ANSWER TYPE

8. $Cr(OH)_3$ and $Al(OH)_3$ cannot be separated using

a) dil. HCl. b) NaOH(excess) c) NH_3 solution d) Any of these

9. Which of the following pairs of cations can be separated from each other by passing H_2S in dil. HCl medium?

a) Sn^{2+}, Hg^{2+} b) Pb^{2+}, Mn^{2+} c) Sb^{2+}, Cu^{2+} d) Zn^{2+}, Cu^{2+}

10. Which of the following compounds will give yellow ppt. On addition of the $(Na_2CrO_4 + AcOH)$ solution?

a) $BaCl_2$ solution b) aqueous suspension of $PbCO_3$

c) aqueous suspension of $BaSO_4$ d) Hot solution of $PbCl_2$

11. Sometimes turbidity appears in the Group II test of group analysis, even in the absence of group II cations. This is due to

a) the presence of Fe^{2+} cation. b) the presence of Fe^{3+} cation.

c) the presence of Zn^{2+} cation. d) the presence of an oxidizing anion.

12. No precipitation is observed when H_2S gas is directly passed through the solution of

a) $ZnCl_2$ b) $NiCl_2$ c) $\left[Ag(NH_3)_2\right]^+$ d) $CoCl_2$

13. Which of the following pairs of cations can be separated by using NaOH solution?

A) Fe^{3+}, Al^{3+} B) Cr^{3+}, Al^{3+} C) Sn^{2+}, Pb^{2+} D) Cu^{2+}, Pb^{2+}

14. A white precipitate is obtained when:

A) a solution of $BaCl_2$ is treated with Na_2SO_3

B) a solution of $NaAlO_2$ is heated with NH_4Cl

C) H_2S is passed through a solution of $ZnSO_4$

D) a solution of $ZnSO_4$ is treated with Na_2CO_3

15. The ion that can be precipitated when $BaCl_2$ solution is added to it is

A) CrO_4^{-2} B) SO_4^{-2} C) Pb^{+2} D) Ag^+

16. How many of the following ions can be detected by the flame test

A) NH_4^+ B) K^+ C) Mg^{2+} D) Ca^{2+}

17. Which of the following hydroxide dissolves in NaOH?

A) $Fe(OH)_3$ B) $Cu(OH)_2$ C) $Al(OH)_3$ D) $Pb(OH)_2$

18. $K_4\left[Fe(CN)_6\right]$ gives the precipitate by the reaction with

A) Fe^{2+} B) Fe^{3+} C) Cu^{2+} D) Zn^{2+}

19. An inorganic salt in its aqueous solution produced a white ppt. with NaOH which dissolves in excess of NaOH. Also its aqueous solution produced light yellow ppt. with $AgNO_3$ sparingly soluble in NH_4OH. The probable salt is

A) $AlBr_3$ B) $BeBr_2$ C) $SnCl_2$ D) $ZnCl_2$

20. A green coloured compound A on heating gives two products B and C. A metal D is deposited by passing hydrogen through heated B. The solution of B in HCl on treatment with potassium ferrocyanide gives brown coloured precipitate. C turns lime water milky. Which of the following cannot be A

A) $CuSO_4$ B) $CuCO_3$ C) $CuCl_2$ D) $Cu(NO_3)_2$

21. $Co^{2+} + KCN$ (not in excess) $\longrightarrow$ precipitate.

Select the correct statement(s) with respect to the precipitate.

A) It is yellow in colour.

B) It is reddish-brown in colour.

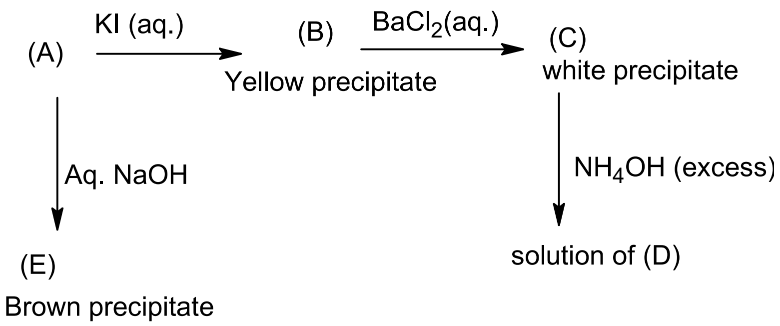
C) It dissolves in excess of the reagent forming a brown solution.

D) It is obtained when brown solution (option (C)) is acidified with dilute HCl in the cold.

22. Potassium ferrocyanide is used to distinguish

A) Cu^{2+} and Zn^{2+} B) Fe^{3+} and Ca^{2+} C) Ag^+ and Zn^{2+} D) Th^{4+} and Cu^{2+}

PASSAGE-1



23. The compound (A) is

A) $CuSO_4$ B) $AgNO_3$ C) $Pb(NO_3)_2$ D) $Ca(NO_3)_2$

24. The compound (D) obtained, when (C) dissolves in excess of NH_4OH will be

A) AgOH B) $\left[Ag(NH_3)_2Cl\right]$ C) Ag_2O D) $AgNO_3$

INTEGER ANSWER TYPE

25. Among the following, the number of cations that give precipitate with Na_2HPO_4 solution is_______.

$Mg^{2+}, K^{+}, Mn^{2+}, Zn^{2+}, Ag^{+}, Fe^{3+}$

26. Among the following, the number of sulphides which are not soluble in yellow ammonium sulphide is______.

$CuS, Bi_2S_3, As_2S_3, SnS_2, CdS$

MATCHING TYPE

27. Match the ions in column-I with the reagents in column-II.

	Column I		Column II
A)	Fe^{3+}, Zn^{2+} & Cu^{2+} can be differentiated by	p)	KI solution
B)	PbS, CuS and CdS dissolve in	q)	Alkaline Na_2SnO_2 solution
C)	Pb^{2+} gives yellow precipitate with	r)	50% HNO_3
D)	Bi^{3+} gives a black precipitate with	s)	K_2CrO_4 solution
E)	$[Ag(NH_3)_2]Cl$ gives back precipitate with	t)	Aqueous NH_3

KEY

1	B	2	C	3	A	4	C	5	A	6	D	7	A	8	ABCD
9	BD	10	ABD	11	BD	12	BD	13	ABD	14	ABCD	15	ABCD	16	BD
17	CD	18	ABCD	19	AB	20	CD	21	BCD	22	ABCD	23	B	24	B
25	5	26	3												

27. (A–T), (B–R), (C–P,S), (D–P,Q), (E–P,R,S)